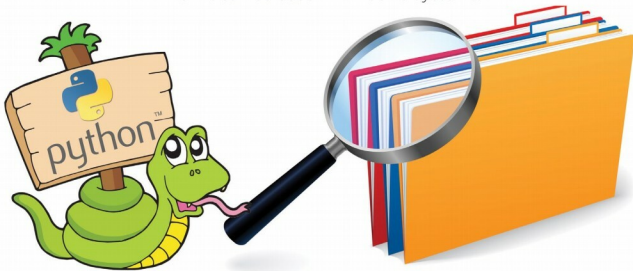


Faster File Search with Python

This article presents a file search utility created by using the power of the versatile Python programming language. Read on to discover how it works and how it can be used in Windows systems.



Computer users often have a problem with file search as they tend to forget the location or path of a file even though Windows provides a file search utility. The Explorer in Windows 7 offers a search facility but it takes around two to three minutes to search a file. In this article, I will give you a Python program which will search a file on your computer's hard disk, within hardly one second.

Let us first understand the program's logic. Figure 1 explains this. Let us first do indexing or, in Python language terms, let's construct a dictionary in which the file will be the key of the dictionary and the value will be the path of the file. The dictionary will be dumped into the *pickle* file. The next time, the file will be searched in the dictionary (dumped in the *pickle* file).

Now that you have understood the logic of the program, let us look at the program in detail. I have broken it into different functions. Let's see what each function does.

```
#program created by mohit
#official website l4wisd.com
# email-id mohitraj.cs@gmail.com
```

The block of code below imports the essential modules:

```
import os
import re
import sys
from threading import Thread
from datetime import datetime
import subprocess
import cPickle
dict1 = {}
```

Next, let's write a function to acquire the drives. This function gets all the drives in your Windows machine. If you have inserted any external/USB pen drive or hard drive disk, the function also obtains details for them.

```
def get_drives():
    response = os.popen("wmic logicaldisk get caption")
    list1 = []
    total_file = []
    t1= datetime.now()
    for line in response.readlines():
        line = line.strip("\n")
        line = line.strip("\r")
        line = line.strip(" ")
        if (line == "Caption" or line == ""):
            continue
        list1.append(line)
    return list1
```

Our next function is the *search1* function, which constructs a dictionary in which the file name is the key and the path is the value of the dictionary.

```
def search1(drive):
    for root, dir, files in os.walk(drive, topdown = True):
        for file in files:
            file= file.lower()
            if file in dict1:
                file = file+"_1"
                dict1[file]= root
            else :
```